

UPSTAGE DOCUMENT AGENT

From notice analysis to eligibility — Moabora

Drop any announcement. The agent reads the brief, cross-checks your profile, and answers: **can you apply?**

JunctionX Korea 2026

Upstage Track

Team Aquarium

01 UNDERSTAND

PDF · HWP · Poster Ingest

02 JUDGE

Automated Eligibility Verdict

03 ACT

Solar Fit Score & Checklist

Notices are everywhere. Eligibility is not.

Students miss vital opportunities not from lack of listings, but from the exhausting cost of manual reading.

PAIN 01

Fragmented Document Formats

- Scattered across PDF guidelines, HWP attachments, image posters, and bulletin HTML.
- Unstructured binary formats that search engines and keyword filters cannot parse.

Bottleneck: Format Incompatibility

PAIN 02

Buried Qualification Clauses

- Crucial constraints ("under 29", "no 4th years", "IT majors only") hidden deep inside multi-page briefs.
- Students must inspect every page just to discover they are disqualified.

Bottleneck: Information Opacity

PAIN 03

Exhausting Reading Overhead

- Cross-checking academic status, age, and major takes 15–30 minutes per notice.
- Evaluating 20+ opportunities requires hours of repetitive, manual toil.

Bottleneck: Time & Cognitive Burden

PAIN 04

Fragmented Channels & Missed Deadlines

- Notices spread across campus portals, Wevity, and ContestKorea without sync.
- Crucial application deadlines pass unnoticed with zero personalized tracking.

Bottleneck: Opportunity Loss

The bottleneck is reading, not listing

Existing aggregators stop at collecting links. The real friction begins after you click.

EXISTING NOTICE BOARDS

List Aggregators (Wevity, ContestKorea)

DOCUMENT ACCESS

Hands you raw attachments. You must download and read HWP/PDFs yourself.

ELIGIBILITY CHECK

100% manual. User spends 15–30 minutes deciphering complex guideline clauses.

RECOMMENDATION

Generic category tags with zero awareness of your actual skills or profile.

Outcome: Manual Workload Pushed to Student

MOABORA DOCUMENT AGENT

Autonomous Document Agent

DOCUMENT ACCESS

Upstage AI reads & structures HWP, PDF, and poster images automatically.

ELIGIBILITY CHECK

Instant 6-rule matching (Eligible Review Ineligible) with explicit reasons.

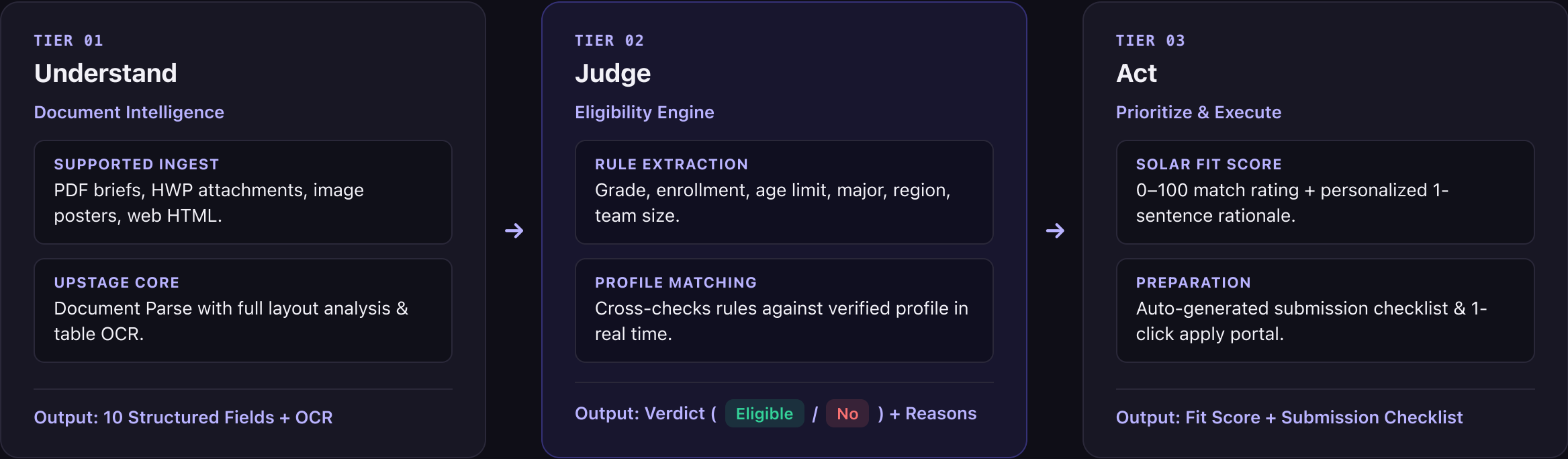
RECOMMENDATION

Solar Pro 4 calculates personalized fit score (0–100) + tailored rationale.

Outcome: Instant Verdict & Submission Checklist

Understand → Judge → Act

A three-tier pipeline transforming raw announcement files into verified application actions.



From zero to application in 4 steps

A frictionless 2-minute workflow from profile setup to finalized application decision.

STEP 01 · 1-MINUTE ONBOARDING

Profile Creation

- Upload certificate (UIE) or paste GitHub / blog URL (Solar Pro 4).
- Auto-extracts university, major, grade, enrollment status, tech stack, and interests.

Deliverable: Verified Student Profile

STEP 02 · MULTI-CHANNEL INGEST

Notice Collection

- Drag & drop PDF / HWP files, paste announcement URLs, or 1-click crawler.
- Automated collection from Wevity and ContestKorea with HWP attachment download.

Deliverable: Multi-Modal Ingestion Pipeline

STEP 03 · PERSONALIZED RANKING

Feed & Fit Scoring

- Feed displays instant eligibility status badges (**Eligible** **Review** **Ineligible**).
- Solar Pro 4 calculates tailored fit score (0–100) with personalized match reasons.

Deliverable: Intelligent Ranked Opportunity Feed

STEP 04 · PREPARATION & ACTION

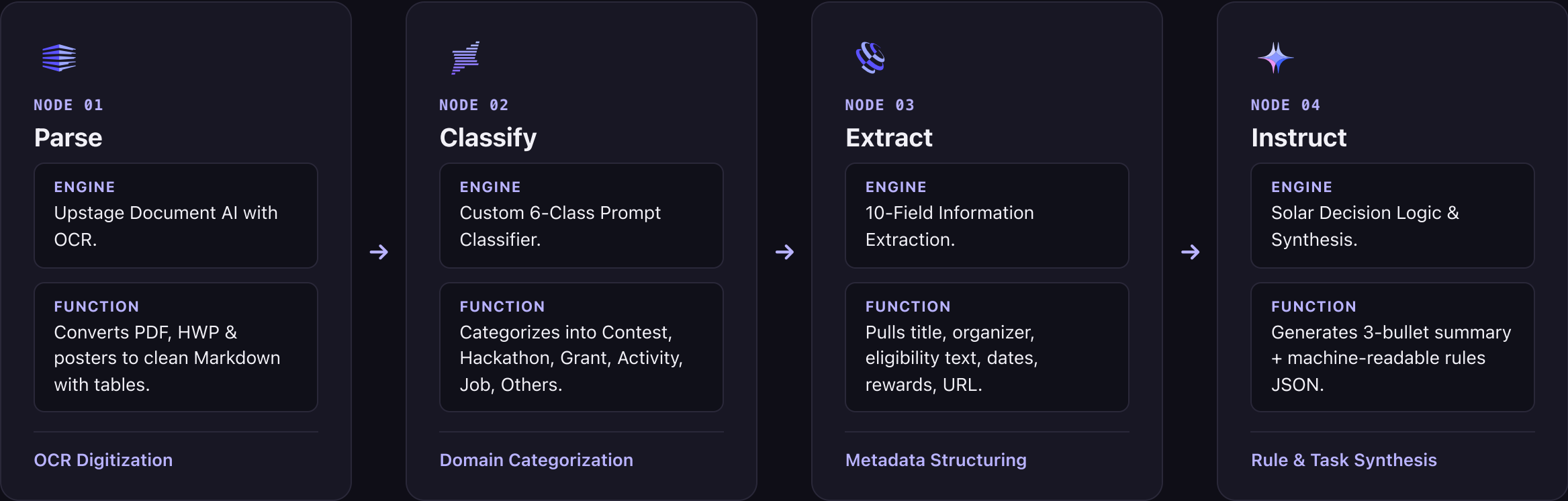
Execution & Doc Chat

- Review per-rule pass/fail explanations, checklist items, and original brief.
- Ask questions and extract custom insights via 3-tool calling Solar Document Chat.

Deliverable: Verified Application Submission

Every notice flows through one 4-node agent

All ingest paths (upload, link, crawler) execute the Studio Agent via REST API with zero bypass.



All four product lines in production

Comprehensive integration across Upstage's Document AI and LLM ecosystem.



Studio Agents

PRIMARY ROLE
Core 4-Node Notice Processing Pipeline (`agt\_...`).

ARCHITECTURE
Accumulating snapshot map ensures full node visibility.

REST /v2/responses



Document Parse

PRIMARY ROLE
High-Precision OCR & Layout Analysis.

INTEGRATION
Powers `parse\_document` tool in Document Chat.

/v1/document-digitization



Information Extract

PRIMARY ROLE
Zero-Shot Schema-Driven Document Extraction.

INTEGRATION
Certificate parsing & `extract\_information` chat tool.

/v1/information-extraction



Solar Pro 4

PRIMARY ROLE
Decision Intelligence & Multi-Tool Orchestration.

INTEGRATION
Calculates 0–100 fit scores + streamed reasoning tokens.

Direct API (OpenAI-Compatible)

Service-grade reliability & transparency

Engineered to handle real-world document edge cases and provide full visibility into AI execution.

ROBUSTNESS

JSON Sanitization

Pre-processes double-encoded Studio Instruct outputs and strips citation markers ([+1]) seamlessly.

Parser Resilience

STATE MANAGEMENT

Snapshot Accumulation

Maintains an accumulating map across Studio polling snapshots so intermediate node outputs are preserved.

Full Node Visibility

TRANSPARENCY

Live NDJSON Streams

Streams pipeline stages, intermediate artifacts, and Solar reasoning tokens in real time to the client.

Streamed Reasoning

RELIABILITY

Resumable Sessions

Server-side workflow sessions survive browser refreshes, preventing duplicate execution and billing.

Refresh-Safe Architecture

OPTIMIZATION

Dual-Layer Caching

SHA-256 document parse cache + profile/notice hash-keyed Solar recommendation cache avoids redundant calls.

Cost & Latency Control

ACCURACY

Deterministic Matcher

Evaluates 6 criteria (grade, status, age, major, region, team) with clear per-rule reason breakdown.

Explainable AI Verdict

Not a list. A definitive verdict.

Transforming manual document reading into automated, transparent intelligence.

TRADITIONAL NOTICE BOARDS

Wevity · ContestKorea

DOCUMENT PROCESSING

Raw HTML and attachments. User must manually download and read HWP files.

ELIGIBILITY VERIFICATION

100% manual check. Students must inspect dense guidelines themselves.

PERSONALIZATION & ACTION

Generic category filters with no user context. User tracks deadlines manually.

Static Link Aggregator

MOABORA DOCUMENT AGENT

Moabora

DOCUMENT PROCESSING

Upstage AI parses HWP, PDF, and posters into structured cards automatically.

ELIGIBILITY VERIFICATION

Instant 6-rule matching with explicit pass/fail rationale (**Eligible** / **No**).

PERSONALIZATION & ACTION

Solar Pro 4 fit scoring (0–100) + auto checklist & 3-tool Document Chat.

Autonomous Decision Engine

3-day hackathon execution & future milestones

Prioritized a rock-solid end-to-end loop while laying an extensible foundation for production scale.

SHIPPED MVP

Delivered in 3 Days

- 3 Ingest paths (Files, URLs, 2-site Crawlers).
- 4-node Studio Agent with full error handling.
- Dual profile extraction (Link & Certificate UIE).
- 6-rule eligibility engine + Solar fit ranking.
- Live NDJSON workflow streams & 3-tool Doc Chat.
- Supabase Cloud storage + Docker deployment.

100% Production Ready

CONSCIOUSLY SCOPED OUT

Deferred Complexity

- **Multi-Tenant Auth:** Single-profile demonstration model.
- **Push Notifications:** Excluded background email/push daemons.
- **Scheduled Crawlers:** Manual trigger-based crawling only.

Avoided Low-Demo Bloat

NEXT MILESTONES

Production Roadmap

- **Multi-User Auth:** Supabase Auth + RLS (schema is RLS-ready).
- **Proactive Alerts:** D-3 deadline notifications via Telegram/Discord.
- **Application Drafts:** Solar-assisted personal statement generator.
- **Team Matching:** Connect students applying to the same contest.

Clear Scaling Path

Four bets we actually shipped

Depth of the stack, a complete loop, a sharper problem, and honest scoping — not a demo shell.

Upstage Technology

- Custom 4-node Studio Agent (``agt_...``) via REST v2 APIs without bypass.
- Production deployment of all 4 Upstage product lines.
- Robust handling of double-encoded JSON and live streaming reasoning.

Max Technical Complexity

Service Completeness

- End-to-end loop: Landing → Profile → Crawl/Ingest → Feed → Detail → Chat.
- Live data from Wevity & ContestKorea with real HWP attachments.
- Resumable workflow sessions that survive browser refresh.

Full Production Loop

Idea Creativity

- Shifting paradigm from "Listing links" to "Delivering application verdicts".
- Reads HWP attachments and posters that traditional aggregators skip.
- Transparent UX displaying live agent thought process and rule proofs.

Novel Problem Framing

Product Planning

- Clear problem focus: Relieving student document fatigue.
- Thoughtful MVP scoping with explicit trade-offs and roadmap.
- Fully documented architecture, data models, and RLS-ready schema.

Execution Rigor

5:30 Live demonstration flow

Step-by-step walkthrough showing end-to-end document intelligence in action.

0:30 – STEP 01

Landing & Concept

Service introduction, student document pain points, and 4-tier pipeline overview.

Architecture Showcase

1:00 – STEP 02

Profile Onboarding

Upload enrollment certificate → Information Extraction creates verified student profile live.

Zero-Shot UIE Live

1:30 – STEP 03

Studio Agent Ingest

Drop hackathon PDF → watch real-time 4-node execution → structured card & instant verdict.

4-Node Agent Stream

0:30 – STEP 04

1-Click Multi-Crawl

Trigger automated collection from Wevity & ContestKorea with HWP attachment extraction.

Real-World Data Ingest

1:00 – STEP 05

Feed & Solar Scoring

Review personalized feed, verdict badges, and Solar Pro 4 fit scores with reasoning.

Personalized Match Feed

1:00 – STEP 06

Detail & Doc Chat

Inspect submission checklist, view original brief, and interact via 3-tool Document Chat.

3-Tool Agent Chat